

Yanwei (Eric) Feng

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PROFESSIONAL SUMMARY

M.S. Industrial & Systems Engineering (USC, 3.82 GPA). Active AI Systems Engineer building production ML and LLM-powered applications at Zhima Program: RAG pipelines, deterministic model output evaluation, Redis-cached data orchestration, and Docker/CI-CD deployment. Background in optimization and statistical modeling applied to feature engineering and constraint-aware system design. Experienced in ML/DL model training and deployment (LoRA fine-tuning, PyTorch, HuggingFace), prompt engineering, and LLM-powered application development over structured financial datasets.

CORE COMPETENCIES

ML/DL Model Training & Deployment

LLM-Powered Application Development

RAG Pipeline Design

Production Data Pipeline Engineering

Prompt Engineering & Model Evaluation

Python · PyTorch · HuggingFace

Statistical Modeling & Feature Engineering

Cross-functional Collaboration

WORK EXPERIENCE

Zhima Program Los Angeles, CA | Dec 2025 - Present

AI Systems Engineer

- Developed and deployed an LLM-powered analytical application with production data pipelines, concurrent orchestration, and confidence-based output evaluation -- serving 4 research scenarios over live financial data
- Designed a deterministic model evaluation framework combining data completeness, signal coherence, and evidence depth metrics to block low-confidence outputs and improve production system reliability
- Fine-tuned Qwen3-0.6B using LoRA on Finance-Alpaca and synthetic financial datasets; built automated data preprocessing, augmentation, and evaluation pipelines for domain-specific model training
- Built RAG pipeline using EDGAR filings, ChromaDB vector search, and sentence-transformer embeddings to ground LLM synthesis in primary-source documents
- Implemented Redis-based caching with workload-specific expiration policies for 4 data sources (market data, news, SEC retrieval, report generation), reducing redundant API calls and improving response latency
- Containerized frontend and backend services using Docker and Docker Compose; established CI/CD validation pipelines with GitHub Actions for automated testing and deployment verification
- Architected production-style FastAPI + Streamlit system with concurrent data orchestration and graceful degradation when external data sources become unavailable

Shelley Whitman Associates (SWA) Los Angeles, CA | June 2025 - Sep 2025

Project Manager

- Managed end-to-end development of Mark's Northeast Furniture e-commerce website, tracking 40+ tasks in OpenProject and Jira to achieve on-time launch
- Facilitated daily cross-functional coordination across web, graphic, and illustration teams; kept deliverables aligned with client requirements and creative vision
- Assessed risk exposure and optimized resource allocation across three concurrent design initiatives, cutting overtime hours by 15%

PROJECTS

Financial Decision Assistant (LLM + LoRA Fine-tuning) Live Demo

Scoped 4 user scenarios (market interpretation, risk, strategy, scenario analysis) from friction points in general-purpose LLM use for trading decisions. Trained and evaluated a fine-tuned Qwen 0.6B model with LoRA for domain-specific financial reasoning; built automated data preprocessing pipeline using Finance-Alpaca. Implemented interactive inference pipeline supporting CLI and batch processing; documented model evaluation results across validation scenarios. Deployed on AWS EC2.

Python · PyTorch · LoRA/PEFT · HuggingFace Transformers · AWS EC2 · Model Evaluation

Scheduling & Timetabling Optimization (Sports & Entertainment)

Modeled scheduling problems with fairness, availability, and resource constraints. Explored integer programming, constraint programming, and local search methods; designed and evaluated solutions for tournament scheduling scenarios. Applied M.S. ISE optimization and quantitative modeling curriculum to real-world constraint-satisfaction problems.

Python · Integer Programming · Constraint Programming · Operations Research

EDUCATION

M.S. Industrial & Systems Engineering University of Southern California (USC) Aug 2024 - June 2026

GPA: 3.82 / 4.0 · Coursework: Optimization & Convex Analysis, Machine Learning, Time Series Analysis, Numerical Methods, Operations Research, Supply Chain Analytics, ERP Systems

B.S. Applied Mathematics University of California, San Diego (UCSD) Sep 2022 - Dec 2023

GPA: 3.6 / 4.0

SKILLS

- ML / DL:** PyTorch, HuggingFace Transformers, Sentence Transformers, scikit-learn, BERT, LSTM, LoRA/PEFT fine-tuning, RAG pipeline design, model training & evaluation, feature engineering
- LLM Systems:** Prompt engineering & evaluation, LLM-powered application development, vector databases (ChromaDB), embedding models, OpenAI API, Anthropic API
- Data & Analytics:** Python (primary), SQL (PostgreSQL, data validation queries), Pandas, statistical modeling, time series analysis (ARIMA), A/B testing concepts, Java
- Backend & Infrastructure:** FastAPI, Streamlit, Redis (workload-specific caching), Docker, Docker Compose, GitHub Actions (CI/CD), AWS EC2, REST APIs
- Tools:** Git, Jira, OpenProject, MS Project, MCP servers, Claude Code, Cursor